nerators must, therefore, be
y regulated to match the active
nd.

nachine speed will vary with
change in frequency which may be
sirable.

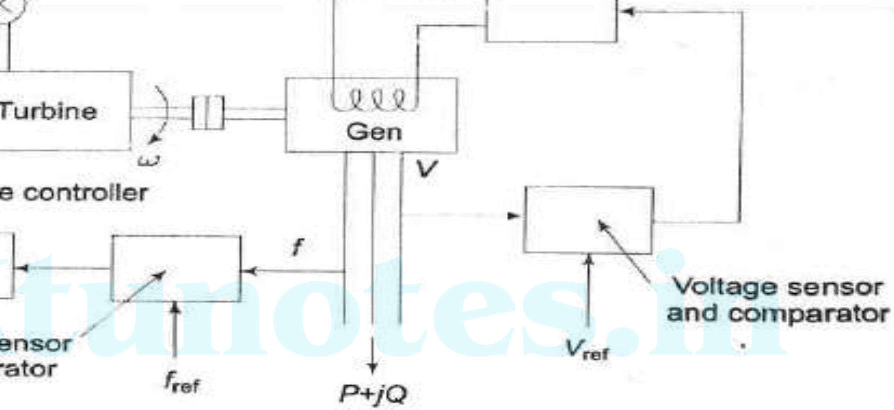
ermissible change in power

± 0.5 Hz)

various system buses may go
prescribed limits.

arge interconnected systems,
lation is not feasible.

eneration and voltage regulation
s installed on each generator.



are set for a particular operating
d they take care of small changes in
without frequency and voltage
e prescribed limits.

hine angle δ and is independent of

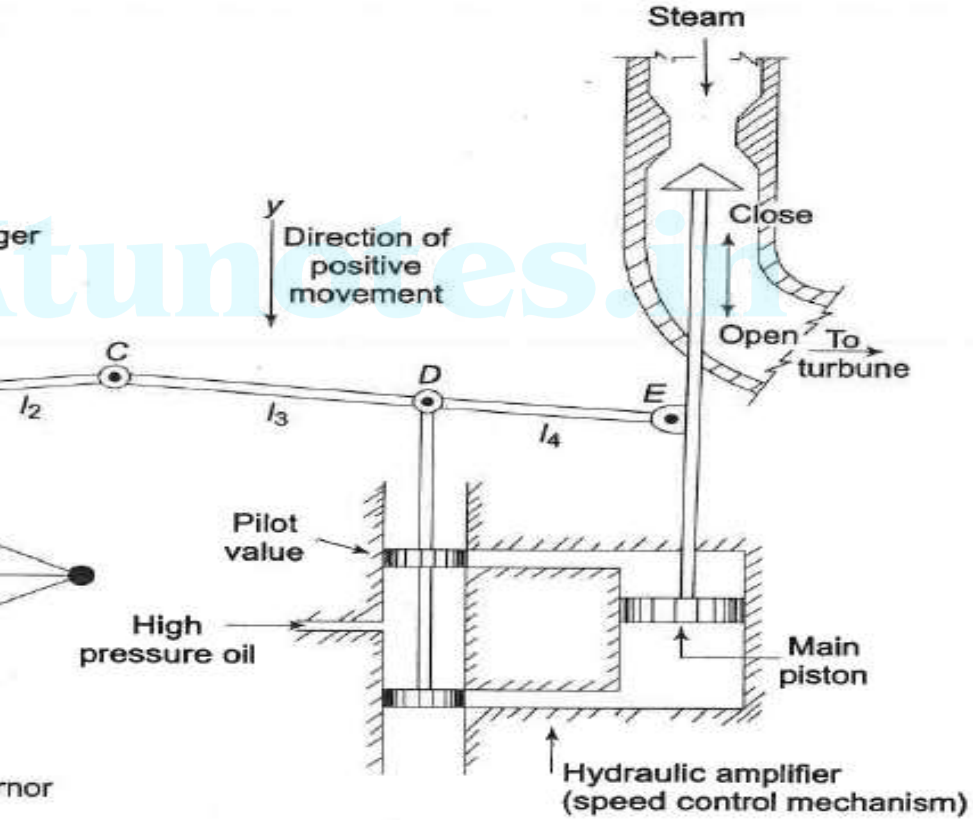
ower generation Q dependent on
e which depends on machine
nd independent on machine angle

be modeled and analyzed
ly.

ontrol is slow acting due to turbine
or moment of inertia .

nsients in excitation voltage
sh much faster and do not affect
s of power frequency control.

ingle turbo generator system
n isolated load.



Steam
Close
Open
To
turbune
ger
y
Direction of
positive
movement
C
D
E
l_2
l_3
l_4
Pilot
value
High
pressure oil
Main
piston
rnor
Hydraulic amplifier
(speed control mechanism)

n speed (frequency).

l increases the fly balls move
d the point B on linkage
moves downwards.

happens when the speed

a pilot valve and main piston

.

level pilot valve movement is
to high power level piston valve

ssary in order to open or close the
against high pressure steam.

gid link pivoted at B and CDE is
d link pivoted at D.

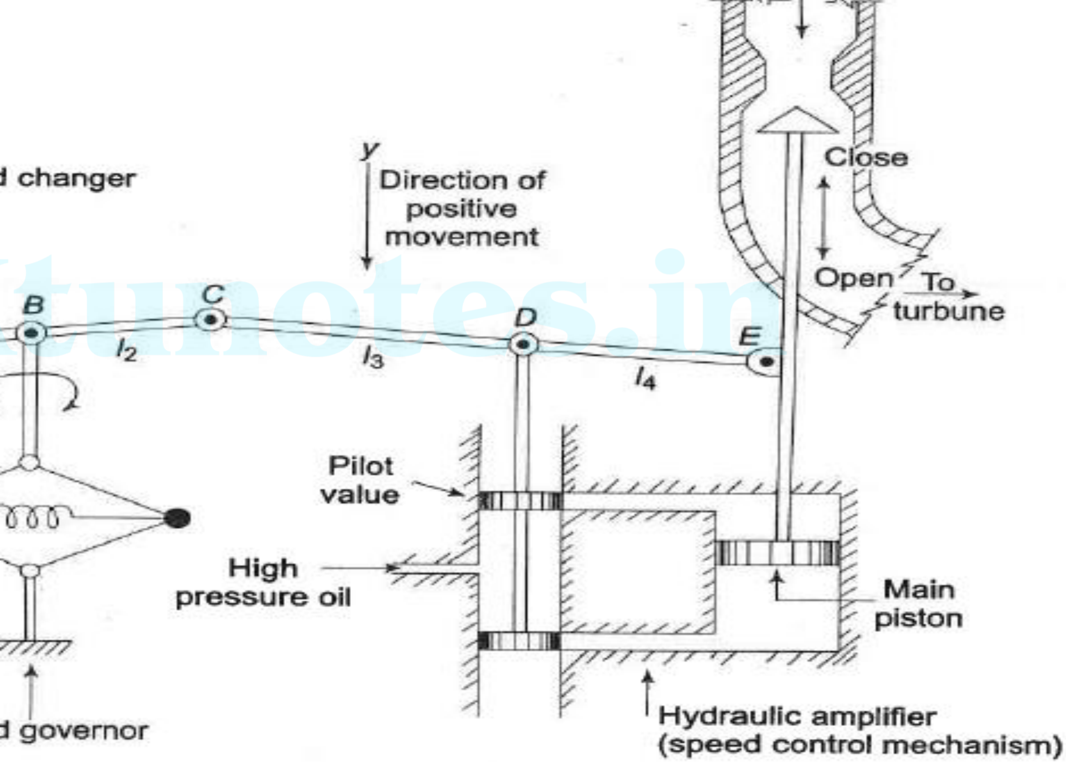
echanism provides a movement to
valve in proportion to change in

des a feedback from the steam
ment.

ady state power output setting for

d movement opens the upper pilot
t more steam is admitted to the
er steady conditions , hence more
r output.

pens for upward movement.


d changer
y
Direction of
positive
movement
Close
Open
To
turbune
B
C
D
E
l_2
l_3
l_4
Pilot
value
High
pressure oil
Main
piston
d governor
Hydraulic amplifier
(speed control mechanism)

y conditions-the linkage

stationary and pilot valve closed,

opened by a definite magnitude,

ing at constant speed with turbine

ut balancing the generator load.

ating conditions

= system frequency (speed)

= generator output = turbine output

= steam valve setting

e is a movement of C, in which two factors

e

ntributes, represented as

$$\Delta y_A = -k_1 \Delta y_A = -k_1 k_c \Delta P_c$$

e in frequency Δf causes the fly balls to

wards so that B moves downwards by a

nal amount $k_2^{'} \Delta f$.

equent movemen t of C with A remaining

y_A *is*

$$_2 \Delta f = +k_2 \Delta f$$

novement of C is therefore

$$k_c \Delta P_c + k_2 \Delta f$$

$$\Big)\Delta y_C + \left(\frac{l_4}{l_3 + l_4}\right)\Delta y_E$$

y_E

t Δy_D, depending upon its sign opens

s of the pilot valve admitting high

o the cylinder thereby moving

pening the steam valve by Δy_E

$)dt$

l from the schematic diagram that a positive
s negative (upward) movement Δy_E accounting
sign used.

elow eqns

$- k_2\Delta f$.......(1)

$_4\Delta y_E$........(2)

$)dt$..........(3)

$$= \frac{k_1 k_3 k_C \Delta P_C(s) - k_2 k_3 \Delta F(s)}{\left(k_4 + \frac{s}{k_5}\right)}$$

$$s) - \frac{1}{R}\Delta F(s)\Big] \times \left(\frac{K_{sg}}{1+T_{sg}s}\right)$$

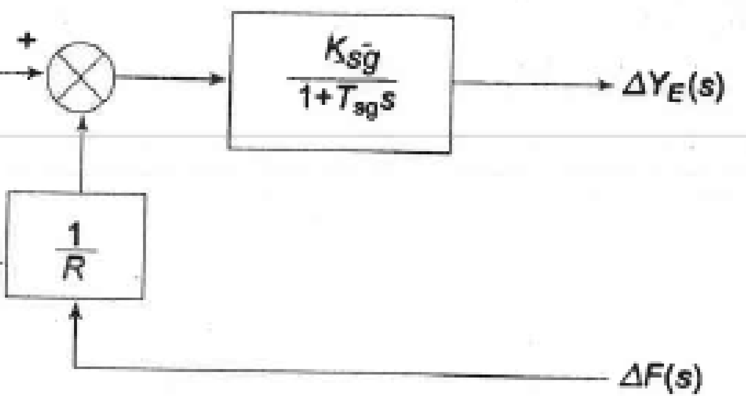
$= \text{speed regulation of the governor}$

$\frac{k_3 k_C}{k_4} = \text{gain of speed governor}$

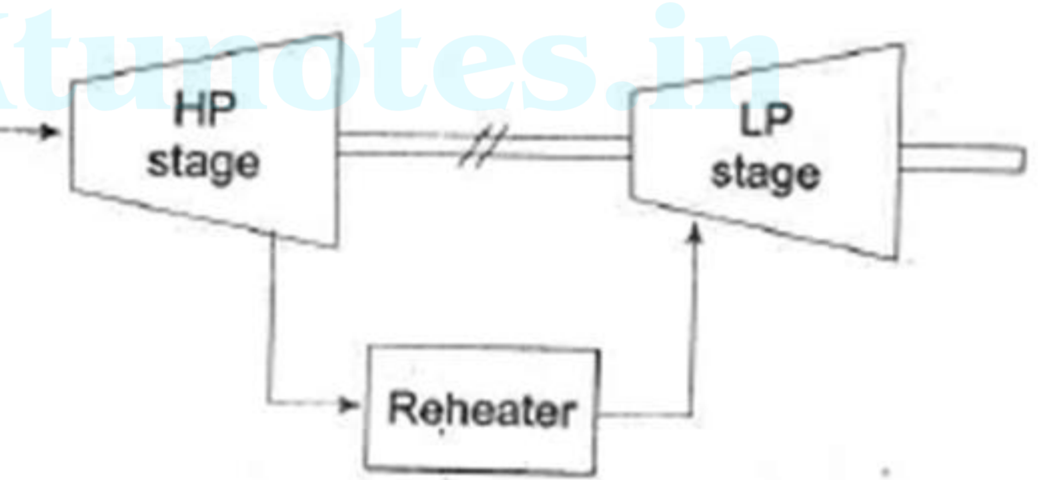
$\frac{}{k_5} = \text{time constant of speed governor}$

$$(s)-\frac{1}{R}\Delta F(s)\Big]\times\left(\frac{K_{sg}}{1+T_{sg}s}\right)$$

n be represented as



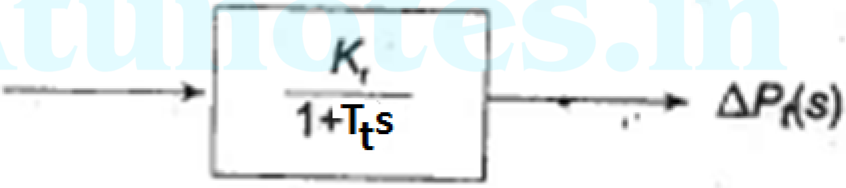
two stage steam turbine with a reheat



ned steam between the inlet steam
rst stage of the turbine

orage action in the reheater which
utput of the low pressures stage to
hat of the high pressure stage

he transfer function model of a steam



ime constant lies in the range of 0.2

incremental turbine power output (assuming generator
e negligible) and ΔP_D is the load increment.

e turbo - generator

eing proportional to square of speed (frequency), the

equency of $(f^0 + \Delta f)$ is given by

^{2}

netic energy is therefore

$\Delta f)$

o frequency, i.e. $\partial P_D / \partial f$ can be regarded

ant for small changes in frequency Δf

ressed as

$B\Delta f$

$$P_D(pu) = \frac{}{f^0}\frac{}{dt}(\Delta f) + B(pu)\Delta f$$

aplace transform, we can write ΔF(s) as

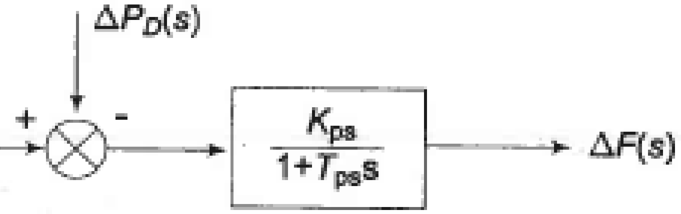
$$\frac{(s) - \Delta P_D(s)}{B + \frac{2H}{f^0}s}$$

$$P_D(s)] \times \left(\frac{K_{ps}}{1 + T_{ps}s} \right)$$

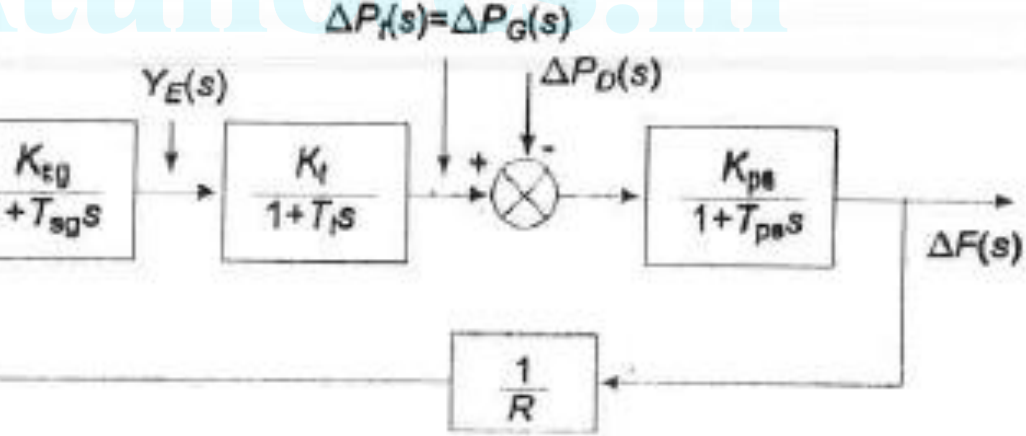
power system time constant \

ower system gain

represented in block diagram form



e block diagrams of Turbine, Generator,
Load.



ator supplying an isolated load.

w a practical system with a number
g stations and loads.

ble to divide an extended power
, national grid) into subareas(may
Electricity Boards) in which the
re tightly coupled together so as to
rent group.

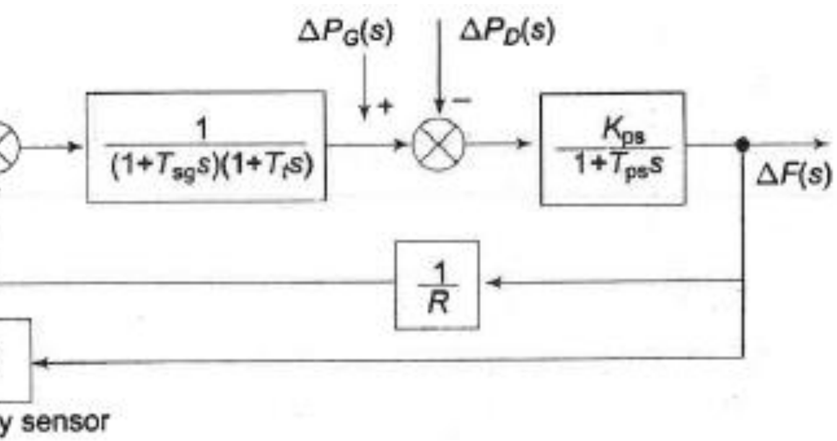
equency is assumed to be the same
in static as well as dynamic

s of developing a suitable control
control area can be reduced to a
d governor, turbo-generator and

trol strategies discussed therefore,
o an independent control area.

y by monitoring the frequency

pose, a signal from Δf is fed through
r to the speed changer.



(s) generated by the integral control

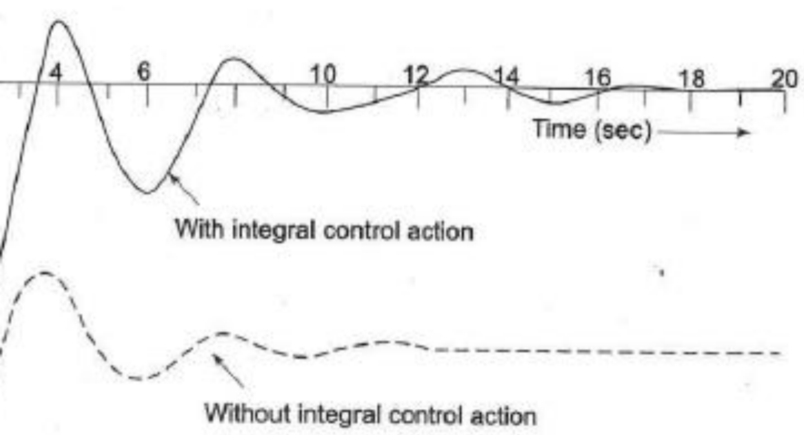
posite sign to ΔF(s) which accounts

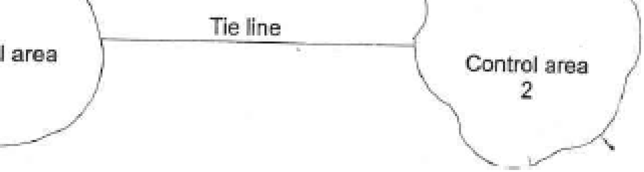
gn in the block for integral controller.

$$\frac{K_{ps}}{{}_{s}s) + \left(\frac{1}{R} + \frac{K_i}{s}\right) \times \frac{K_{ps}}{(1+T_{sg}s)(1+T_t s)}} \times \frac{\Delta P_D}{s}$$

$$\frac{RK_{ps}s(1+T_{sg}s)(1+T_t s)}{1+T_t s)(1+T_{ps}s)R + K_{ps}(K_i R + s)} \times \frac{\Delta P_D}{s}$$

ad frequency control of a given
the change (error) in frequency is
rea Control Error (ACE).





bjective now is to regulate the each area and to simultaneously ie line power as per inter-area power

e of frequency, proportional plus roller will be installed so as to give tate error in tie line power flow as the contracted power.

d with suffix 1 refer to area-1 and ffix 2 refer to area-2

l control area case the incremental $P_D)$ was accounted for by the rate of ored kinetic energy and increase in sed by increase in frequency.

ne transports, power in or out of an t must be accounted for in the power balance equation of each area.

ngles of equivalent machines of the

as

changes in δ_1 *and* δ_2, the incremental

n be expressed as

$$\Delta\delta_1 - \Delta\delta_2)$$

$$\frac{\;\;_2|}{\;\;_{12}} \cos\left(\delta_1^0 - \delta_2^0\right) = \text{synchronizing coefficient}$$

$$dt - \int \Delta f_2 dt \Big)$$

re incremental frequency changes of areas 1

ental tie line power out of area 2 is given by

$$dt - \int \Delta f_1 dt \Big)$$

$$\cos\left(\delta_2^0 - \delta_1^0\right) = \frac{P_{r1}}{P_{r2}} T_{12} = a_{12} T_{12}$$

noted that all quantities other than

are in per unit

e Laplace transform

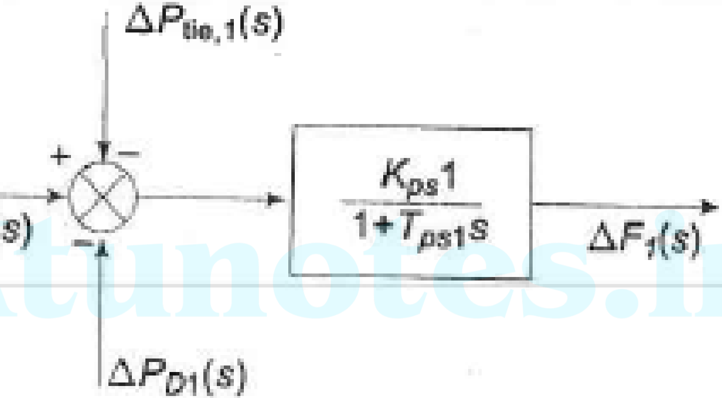
$$\Delta P_{G1}(s) - \Delta P_{D1}(s) - \Delta P_{tie,1}(s)\Big] \times \frac{K_{ps1}}{1 + T_{ps1}s}$$

$\overline{\quad}_{0}$

to the isolated control area case,

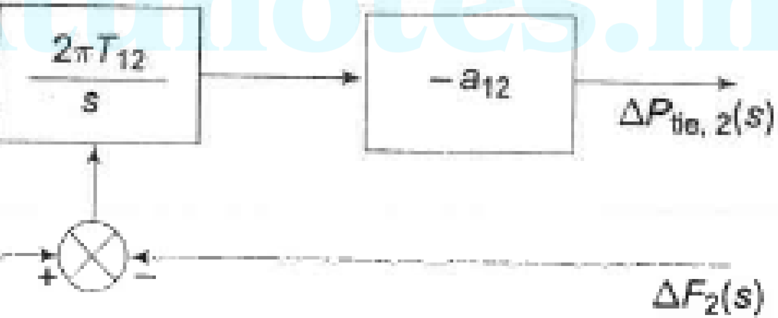
hange is the appearance of the signal

$$\frac{2\prod T_{12}}{s}\left[\Delta F_1(s) - \Delta F_2(s)\right]$$



$\Delta P_{tie,1}(s)$
+
−
−
s)
$\frac{K_{ps1}}{1+T_{ps1}s}$
$\Delta F_1(s)$
$\Delta P_{D1}(s)$

$$\frac{\quad}{s}\left[\Delta F_1'(s) - \Delta F_2'(s)\right]$$



rol loop forced the steady state
ror to zero.

ate tie line power error in a two-area ade to zero by another integral (one for each area)

tes the incremental tie line power ed it back to speed changer.

nplished by a single integrating block g ACE as a linear combination of frequency and tie line power.

ace transform

$\Delta P_{tie,1}(s)+b_1\Delta F_1(s)$

r the control are a 2, ACE_2 is expressed as

$\Delta P_{tie,2}(s)+b_2\Delta F_2(s)$

the basic block diagrams of the two control areas

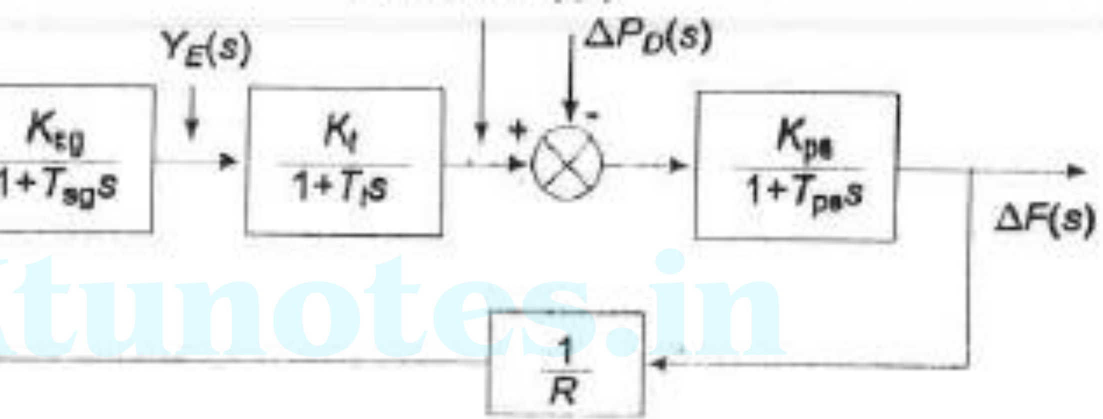
ng with $\Delta P_{c1}(s)$ and $\Delta P_{c2}(s)$ generated by integrals of

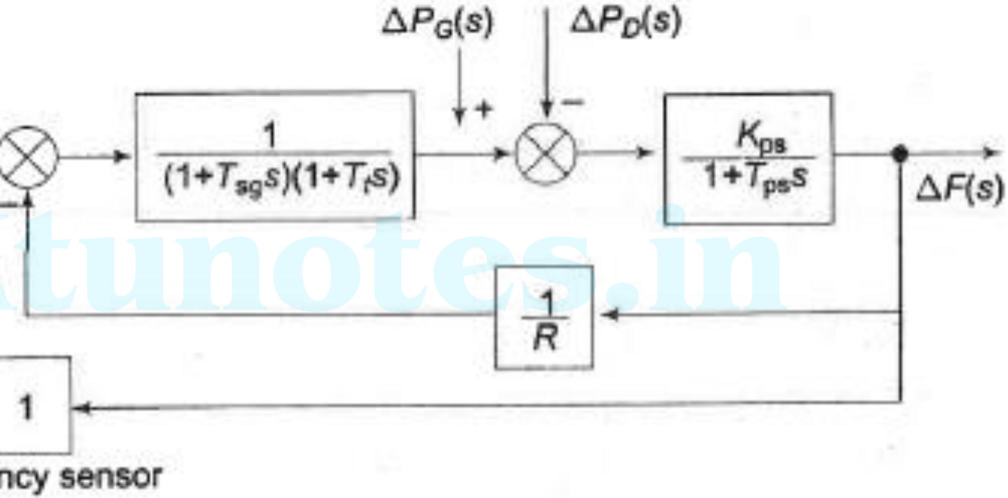
CEs (obtained through signals representing changes

wer and local frequency bias)

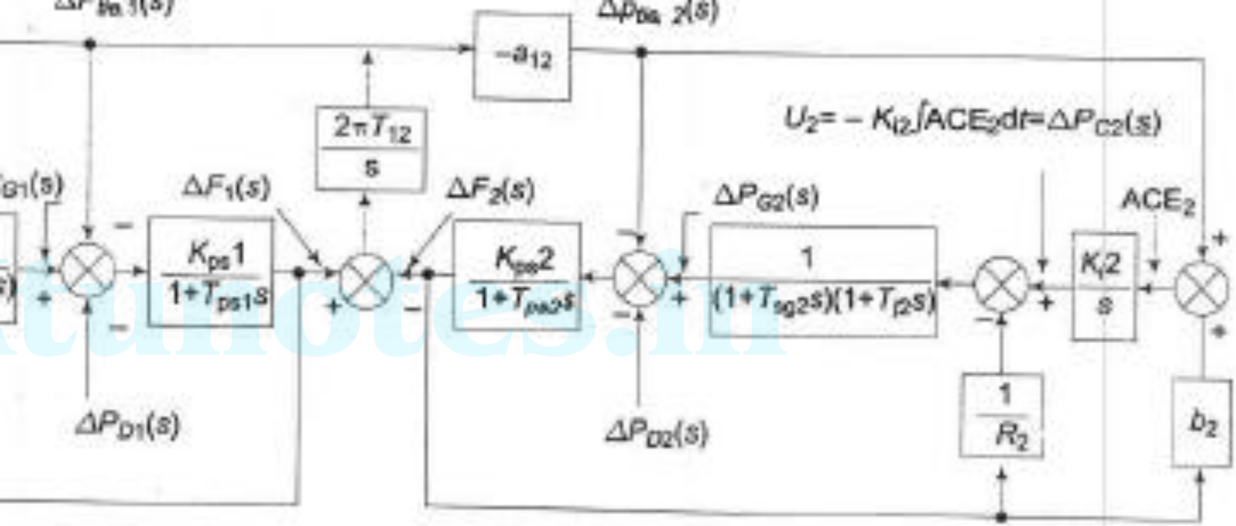
changesin loads ΔP_{D1}, and ΔP_{D2} be simultaneously

ontrol areas 1 and 2, respectively.

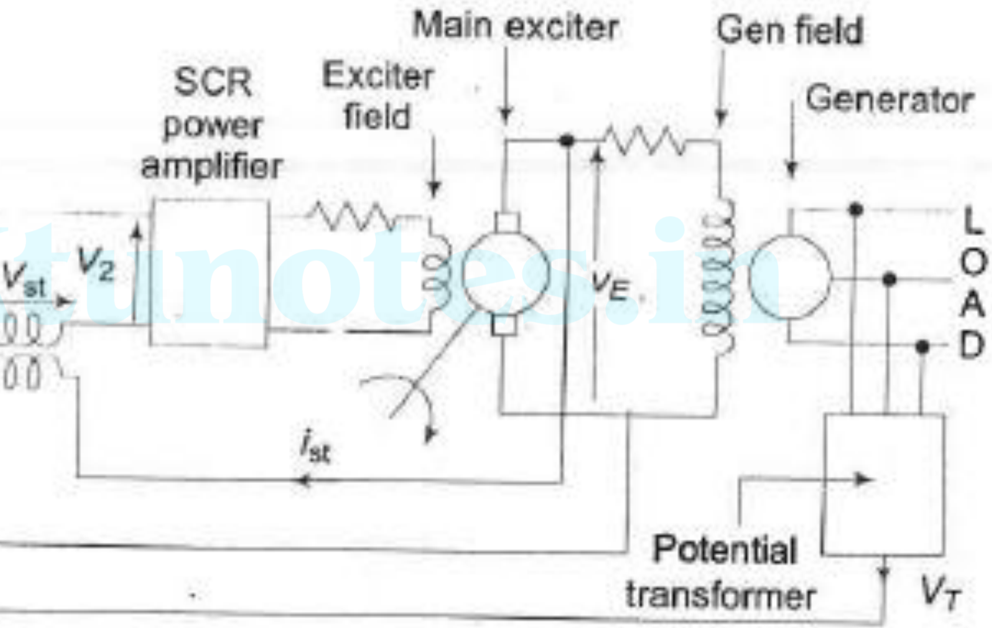

$Y_E(s)$
$\Delta P_D(s)$
$\frac{K_{sg}}{1+T_{sg}s}$
$\frac{K_t}{1+T_ts}$
+
-
$\frac{K_{ps}}{1+T_{ps}s}$
$\Delta F(s)$
$\frac{1}{R}$



ΔP_G(s)
ΔP_D(s)
1
(1+T_sg s)(1+T_t s)
K_ps
1+T_ps s
ΔF(s)
1
R
1
ncy sensor



site block diagram of two-area load frequency control (feedback
provided with integral of respective area control errors)



Main exciter
Gen field
SCR
power
amplifier
Exciter
field
Generator
V_{st}
V_2
v_E
i_{st}
L
O
A
D
Potential
transformer
V_T

field is automatically controlled
r $e = V_{ref} - V_T$, suitably amplified
age and power amplifiers.

e

ating error, $e = V_{Ref} - V_T$

es the corrective action of adjusting the
ation.

and amplifies the error signal.

fier and exciter field:

ary power amplification to the signal for
exciter field.

ter field time constant.

d by the main exciter voltage V_E.

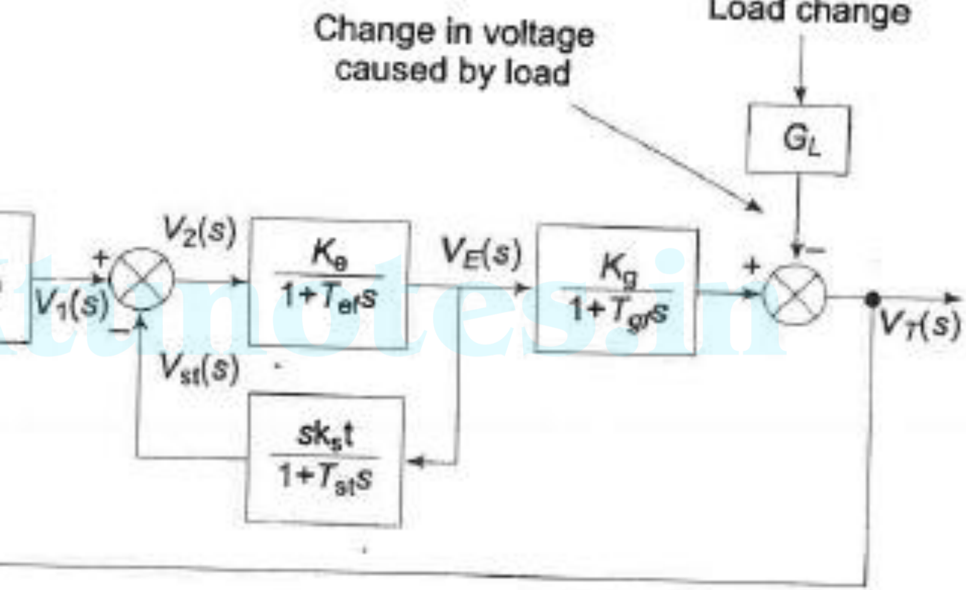
produces a voltage proportional to field current.

sfer function is

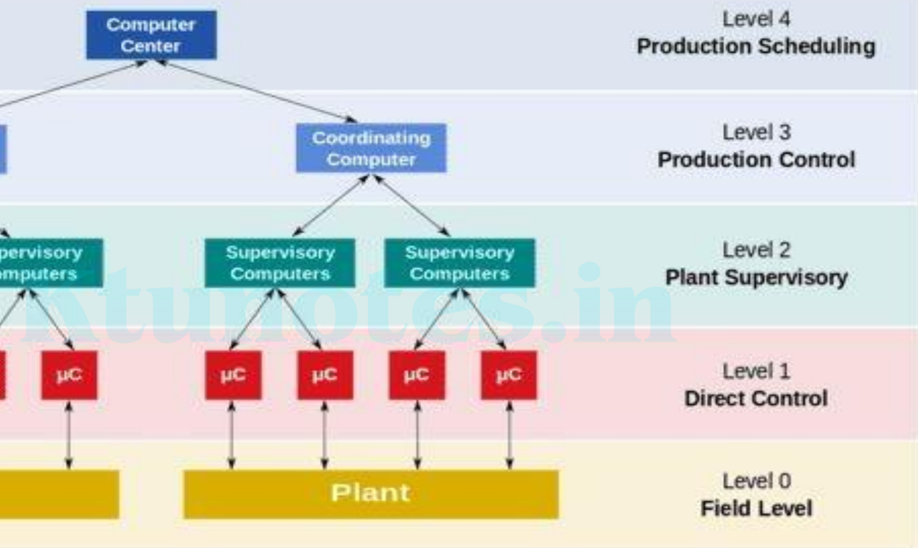
erator field time constant

gh time constants to impair the system's dynamic response.

ynamic response of a control system can be improved by the ck loop. The derivative feedback in this system is provided by ansformer excited by the exciter output voltage V_E. The output rmer is fed negatively at the input terminals of the SCR power nction of the stabilizing transformer is



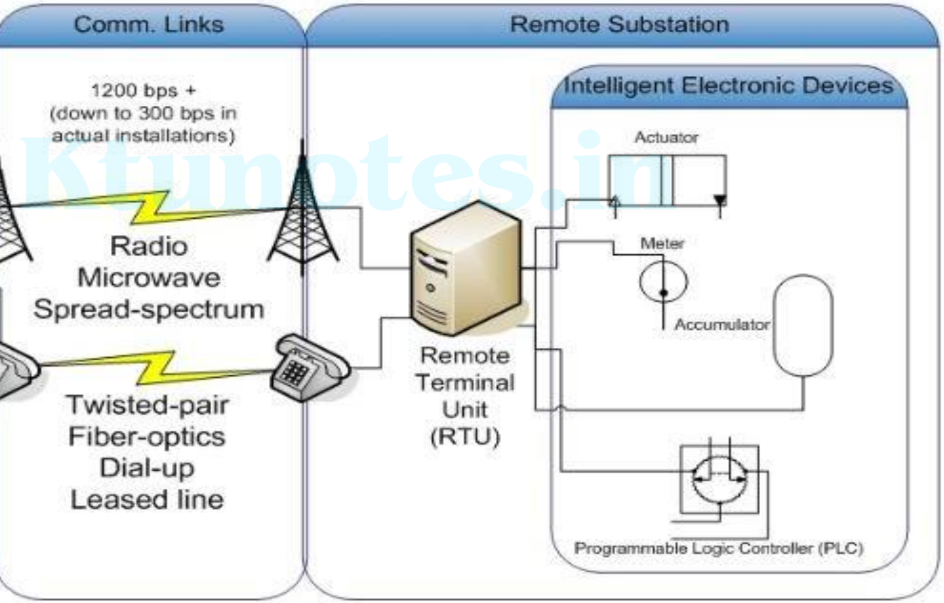
k diagram of alternator voltage regulator scheme



neral term for a high-level of overall control of many individual rol loops. It gives the operations supervisor an overview of the ntegration of operation between low-level controllers.

n be processed by computing machines.

CADA system are mentioned below.



ol

DA refers to sending command messages to a device to operate the and Controls system (I&C) and power-system devices. Conventionally, human managers to initiate command from an operator console on the r. Field personnel can also control machines using front panels.

ollection

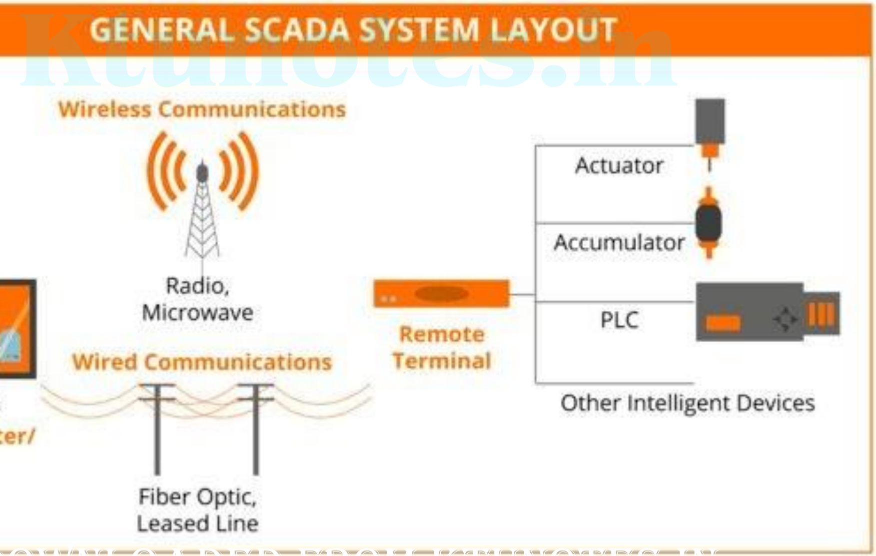
cting data and filling datasheets by hand, SCADA automatically compiles eal-time. SCADA gathers data from hundreds or even thousands of sensors t also generates backlogs for later analysis.

unication:

information to a central hub. A communication network transport all the rom sensors. Earlier systems had radio or modem. Today, SCADA data is internet protocol (IP) and Ethernet.

ents and its functions

ses of the following components



various locations. The more complex a system, the more sensors we may

units:

onsible for collecting data, but we also need something to be able to oret the data. This is where the conversion unit comes in. Conversion units ized units deployed at a specific location in the field. These are connected convert the information they receive into digital format, which is then sent system to display. The two most common types of conversion units used m are PLCs and RTUs. How do we determine which unit we need? that our specification is.

able logic controllers (PLCs)

gic controllers are good for situations where we want more localized rammable logic controller is an industrial digital computer designed for ents and multiple inputs. PLC is used sometimes in place of other due to their versatility, flexibility, affordability, and configuration. However, od programming skills to make the most out of it.

nctions covering a broad area geographically.

units assist as local collection points for gathering information from commands to control and protection relays.

network:

exist without a properly designed communication network system. All ects rely solely on the communication network. It provides a channel etween the supervisory control, the data acquisition units, and any ected to the system. The main function of a communication network n is to connect the Conversion units with the SCADA master station. smitted through various communication platforms such as ethernet, line carrier communication, optical fiber line, cellular, radio, satellite, other wireless protocol. Most facilities have specialized integrated eld buses, wired or wireless, due to security reasons.

considered to be the supervisory computer system because they serve as centralized processing unit. Although the units themselves are typically onsoles, there are several other SCADA components, such as software s, which could be named under this category as well.

g, Human Machine Interface (HMI) is a user interface or dashboard that n to a machine, system, or device. In a SCADA system, it allows the and interact with collected and processed data. This interface is usually tasks like collecting data, creating maps, diagrams, sending out making reports.

munication server (RCS):

nd analyzing the data gathered in the SCADA system, one needs a digitally ure place to store this vast database. The human-computer interface (HCI) y requests data from a server responsible for data acquisition, a dware that is used to connect software services to the conversion units e server makes data acquisition from these local units possible.

uninterrupted power supply and a protected environment

ow the status of a complex power system in real-time

onitor and control system that are in remote areas

nsmission and distribution sectors, supervision, monitoring, and ects in all these areas. Therefore, the SCADA implementation of e overall efficiency of the system for optimizing, supervising, and n, transmission & distribution systems. SCADA function in the ffers greater system reliability and stability for integrated grid perations including protection, monitoring, and controlling. To provide minimize operational costs, and to preserve capital investment, the en as seriously in generating stations.

tions of SCADA in power plants include:

nuous monitoring of speed and frequency of electrical machines

aphical monitoring of coal delivery and water treatment processes

city generation operations planning

ol of active and reactive power

and turbine protection and their condition in case of thermal plant

oring of renewable energy farms and load dispatch planning

scheduling

ical data processing of all generation related parameters nce of events recording

e auxiliary systems and supporting components needed to run the nd deliver energy. SCADA system is also highly effective in the e of Plant.

ansmission system:

oonding circuit model parameters are often in error as compared he SCADA system. Without a SCADA system, these errors cause the erroneous, and hence, lead to increased costs or incorrect billing. affect state estimator analysis, contingency analysis, short circuit g, machine stability calculations, and transmission planning in case . SCADA integration into the transmission system is significantly relay interface/interaction

gulation management

hanger control

er management

modeling

circuit isolation control and interactive switch control display

eal-time single-line displays

eration and maintenance logs

system diagnostics by using system-defined controller alarms nagement)

CADA implementation to the power distribution not only reduces the
peration and its cost but facilitates smooth processes by reducing

athers the data from various electrical substations and correspondingly in substations continuously monitor the substation components and ansmits that to the central system.

proving efficiency by maintaining a tolerable range of power factor

niting peak power demand

nding and alarming the operators by identifying the problem spot

torian data and viewing that from remote and barely inaccessible locations

ick response to customer service interruptions

eder automation and Load Sectionalizer

vide the ability to over-ride automatic control of capacitor banks

ial and abnormal conditions which may affect the quality like
distortions

es of SCADA

tremely advantageous way to run and monitor processes. They are
ons, such as climate control. They can also be effectively used in
s monitoring and controlling a nuclear power plant or mass transit

nance:

e errors by accurately measuring data and increasing the overall

istness:

t of SCADA is performed within a well-established framework that
obustness where power requirement is crucial.

he quality of the produced electric energy profile using standard

nd maintenance costs:

are required to monitor field gear in remote locations, this reduces
g costs.

iness systems:

easily integrated with the business systems, leading to increased
ity.

emented on a large scale in power systems so as to increase their nd durability. Data acquisition and monitoring can be very power systems are upgraded to SCADA. Now, electrical systems l intelligent to monitor and control all of the involved operations become possible only because of technological advancements. s essential for the power sector to optimize their systems as per chnical changes.

ver system analysis-
merical Problems

$$\Delta f = (-\Delta P_L)\,\frac{1}{D + \frac{1}{R}}$$

quency sensitive load co-efficient

peed regulation

oad change in p.u.

system

$$f = \frac{-\Delta P_L}{\left(D_1 + \frac{1}{R_1}\right) + \left(D_2 + \frac{1}{R_2}\right)}$$

$$\Delta P_{m2} = \frac{-\Delta f}{R_2}$$

er flow

$$\Delta P_{12} = -\Delta P_{21} = \Delta f\left(\frac{1}{R_2} + D_2\right)$$

ator and its control have following parameters

constant = 5sec

nstant τ_g = 0.25sec

tant τ_T = 0.6sec

egulation = 0.05 pu

output is 200 MW at 50 Hz. The load suddenly increases by

state frequency deviation.

$= 0.25$ pu

$$\frac{\;_L}{\frac{1}{R}} = \frac{-0.25}{0.8 + \frac{1}{0.05}} = -0.01202 \text{ pu.}$$

cy deviation $\Delta f_{ss} = -0.01202 \times 50 = -0.601$ Hz.

Area	1	2
egulation	0.05	0.0625
ncy sensitive load co-efficient	0.6	0.9
constant	5	4
or time constant	0.2	0.3
e time constant	0.5	0.6

g in parallel at a nominal frequency of 60 Hz. The co-efficient is given as 2.0 pu. If the load in area1 increases by the new steady state frequency and the change in tie-line flow.

is

$$= 0.1875\,\text{p.u}$$

y deviation is

$$\frac{-\Delta P_{L1}}{\frac{1}{R_2} + D_1 + D_2} = \frac{-0.1875}{20 + 16 + 0.6 + 0.9} = -0.005\,\text{p.u}$$

cy deviation in Hz is

$$\Delta f = (-0.005)(60) = -0.3\,\text{Hz}$$

and the new frequency is

$$f = f_0 + \Delta f = 60 + (-0.3) = 59.7\,\text{Hz}$$

$$\frac{1}{R_2} + D_2 \Big)$$

$$05\left(\frac{1}{0.0625} + 0.9\right)$$

$$05 \times 16.9 = -0.0845\, pu$$

$$5\ MW$$

ange in mechanical power in each area is

$$_1 = -\frac{\Delta\omega}{R_1} = -\frac{-0.005}{0.05} = 0.1\,p.u$$

$$= 100MW$$

$$_2 = -\frac{\Delta\omega}{R_2} = -\frac{-0.005}{0.0625} = 0.08\,p.u$$

$$= 80MW$$

= 0.6; $H_1 = 4.5$;

= 0.85; $H_2 = 5.0$;

nning in parallel at a frequency of 50 Hz. The synchronizing

pu at the initial operating angle. A load change of 150 MW

nine the new steady state frequency and the change in tie–line

$$\frac{}{1000} = 0.15 \text{ pu.}$$

equency deviation is

$$\frac{-\Delta P_{L1}}{\quad_1 + \beta_2}$$

$$+ D_1 = \frac{1}{0.045} + 0.6 = 22.822$$

$$+ D_2 = \frac{1}{0.06} + 0.85 = 17.516$$

$$\frac{-0.15}{2.822 + 17.516} = -0.0037 \text{ pu}$$

state frequency = 50 − (0.0037 × 50) = 49.815 Hz.

$0037 \times 17.516 = -0.0648 \text{ pu} = -64.8 \text{ MW}.$

ve, it implies that 64.8 MW flows from area 2 to area 1.

cal powers is given by

$$\frac{0.0037}{0.045}\Bigg) = 0.082 \text{ pu} = 82 \text{ MW}$$

$$\frac{0.0037}{0.06}\Bigg) = 0.0617 \text{ pu} = 61.7 \text{ MW}$$

0 MVA 50 Hz turboalternator operates at no load at 3000 r.p.m. A pplied to the machine and the steam valves to the turbine commence the time-lag in the governor system. Assuming inertia constant H enerator capacity, calculate the frequency to which the generated eam flow commences to increase to meet the new load.

ion $H = \dfrac{\text{Stored energy}}{\text{Capacity of machine}}$

d at no load = $4.5 \times 100 = 450$ MJ

ves open the energy lost by the rotor = $25 \times 0.6 = 15$ MJ

here is reduction in speed of the rotor and, therefore, reduction in

) the new frequency will be

$$f_{new} = \sqrt{\frac{450 - 15}{450}} \times 50 \text{ Hz} = 49.16 \text{ Hz.} \quad \textbf{Ans.}$$